

Md. Ittesaf Hossain

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https://portfolio-five-navy-1pk621p34c.vercel.app/

Professional Summary

Junior AI/ML engineer with 8 months of experience as an AI Developer Trainee and research experience in medical imaging and satellite change detection. Lead author of two IEEE conference papers on explainable ensemble learning. Builds and ships models with PyTorch and FastAPI, from experiment to deployed application.

Experience

AI Developer Trainee

Sparktech AI

Feb 2026 – Present

Mohakhali, Dhaka, Bangladesh

- Built LLM agent and chatbot features with LangGraph/LangChain and the OpenAI and Gemini APIs, including prompt engineering and tool integration, and delivered them to clients in production.
- Developed FastAPI backend services and integrations that expose RAG-based document Q&A and other AI features to client applications.
- Trained and evaluated machine-learning and computer-vision models in PyTorch and TensorFlow; owned features end to end from design to deployment.

Research Assistant / Student Researcher

Visual Data Analysis Lab (VDAL), Green University of Bangladesh

Oct 2025 – Present

Dhaka, Bangladesh

- Thesis: bi-temporal building damage assessment on the xView2 dataset, using a ResNet-152 backbone with a BIT transformer encoder to segment damage from pre- and post-disaster satellite image pairs.
- Authored or co-authored five research manuscripts (two published at IEEE conferences) on explainable stacking ensembles and CNN ensembles with SHAP, LIME and Grad-CAM++.

Education

Green University of Bangladesh

B.Sc. in Computer Science and Engineering

CGPA: 2.97/4.00 (final-semester result pending)

2022 – Sep 2026

Dhaka, Bangladesh

Projects

Building Damage Assessment (Thesis) | PyTorch, BIT Transformer, ResNet-152, xView2

github.com/IH-Arik/CNN-Based-Damage-Change-Segmentation

- Designed a bi-temporal change-detection model (6-layer, 8-head BIT transformer encoder with multi-scale feature fusion) to produce pixel-wise damage masks.
- Trained for 100 epochs with rotation, flip, colour-jitter, noise, cropping and mixup augmentation.

Brain Tumor Classification with RAG | PyTorch, FAISS, Flask, Docker

github.com/IH-Arik/brain-tumor-rag

- Fine-tuned a ResNet18 classifier for four MRI classes (glioma, meningioma, pituitary, no tumor) and served it through Flask endpoints.
- Built a retrieval pipeline (all-MiniLM-L6-v2 embeddings, FAISS index over 10+ medical documents) so every answer returns its source documents.

GoCustify AI (Mabdel) | FastAPI, MongoDB, LangGraph, React Native

github.com/IH-Arik/Mabdel

- Built a FastAPI backend with async MongoDB and JWT/OTP authentication for an AI business-communication platform with React web, React Native mobile and admin apps.
- Integrated GPT-4o (reply suggestions, document generation), Whisper (call transcription) and LangGraph (AI command routing); wrote API tests with Pytest.

PPE and Hazard Detection | YOLO, TFLite, Streamlit, React Native

github.com/IH-Arik/PPE-Detection

- Built YOLO-based detection of helmets, vests, masks, boots, gloves and glasses (and their absence) in images, video and live camera feeds.
- Shipped it as a Streamlit web app and an on-device React Native app using TFLite export.

Publications

Md. Ittesaf Hossain, Md. Rajibul Palas, Mozdaher Abdul Quader, Sabbir Hosen Mamun, Md. Sharifur Rahman.

“Explainable Ensemble Modeling for Diabetes Risk Prediction Using the Real-World DiaBD Cohort.” *IEEE*

International Conference on Quantum Photonics, Artificial Intelligence and Networking (QPAIN), Apr 2026. DOI:

10.1109/QPAIN69676.2026.11545949

Md. Ittesaf Hossain et al. “Explainable Ensemble Machine Learning Framework for Accurate Detection of Thyroid Disorders.” *International Conference on Sustainable Technology and Engineering (i-COSTE)*, *IEEE*, Dec 2025. DOI: 10.1109/i-COSTE68047.2025.11467458

Md. Mustakim Hossain, **Md. Ittesaf Hossain**, Nur Hossain. “RDWS-EKSD: ROI-Guided Dynamic Weighted Stacked Ensemble with Grad-CAM++ for Explainable Kidney Stone Detection.” Preprint.

Md. Ittesaf Hossain, Tanha Tabassum Tusmi, Mayeesha Farjana. “An Adaptive and Intelligent Ensemble Framework for Efficient Sudoku Solving: Integrating Classical CSP Algorithms.” Preprint.

Technical Skills

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, LightGBM, CatBoost, CNNs, Vision Transformers, stacking ensembles, SHAP, LIME, Grad-CAM++, NumPy, Pandas

Computer Vision: OpenCV, YOLO (Ultralytics), change detection, medical imaging, ONNX, TFLite

LLMs and RAG: LangGraph, RAG (FAISS, sentence-transformers), OpenAI API (GPT-4o, Whisper), Gemini API, Hugging Face

Backend: Python, FastAPI, Flask, Node.js, Express, MongoDB, PostgreSQL, Docker

Frontend and Mobile: React, Next.js, TypeScript, JavaScript, Tailwind CSS, React Native, Flutter

Tools: Git, GitHub, Linux, Jupyter, Pytest, Vercel